

Mobile Application Development Lab Record

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Lab 1: BASICS OF ANDROID MOBILE APPLICATION DEVELOPMENT TOOL

Q1: Create an Android application to show the demo of displaying text with justifying elements, changing text colors ,fonts etc.

MainActivity.java:

```
package com.example.demo;

import android.graphics.Color;
import android.graphics.Typeface;
import android.os.Bundle;
import android.view.View;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView textView;
    float currentSize = 18f;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        textView = findViewById(R.id.textview);
    }

    public void makeRed(View view) {
```

```

        textView.setTextColor(Color.RED);
    }

    public void makeBlue(View view) {
        textView.setTextColor(Color.BLUE);
    }

    public void makeBold(View view) {
        textView.setTypeface(null, Typeface.BOLD);
    }

    public void makeItalic(View view) {
        textView.setTypeface(null, Typeface.ITALIC);
    }

    public void increaseSize(View view) {
        currentSize += 2;
        textView.setTextSize(currentSize);
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Demo Text"
        android:textSize="18sp"
        android:padding="8dp"
        android:justificationMode="inter_word"/>

    <Button
        android:id="@+id/redBtn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Make Text Red"

```

```
android:onClick="makeRed"/>
```

```
<Button  
    android:id="@+id/blueBtn"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Make Text Blue"  
    android:onClick="makeBlue"/>
```

```
<Button  
    android:id="@+id/boldBtn"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Bold Text"  
    android:onClick="makeBold"/>
```

```
<Button  
    android:id="@+id/italicBtn"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Italic Text"  
    android:onClick="makeItalic"/>
```

```
<Button  
    android:id="@+id/increaseSizeBtn"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Increase Text Size"  
    android:onClick="increaseSize"/>
```

```
</LinearLayout>
```

UI:



After changing the text size and the text color:

Demo Text

Make Text Red

Make Text Blue

Bold Text

Italic Text

Increase Text Size

Q2: Find the “hello word” text in the XML document and modify the text.

MainActivity.java:

```
package com.example.demo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main.xml:

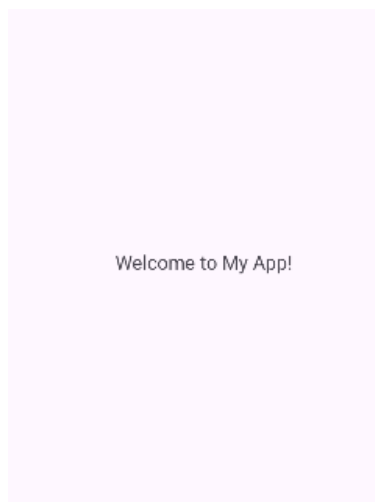
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="148dp"
        android:layout_marginTop="248dp"
        android:layout_marginEnd="117dp"
```

```
        android:layout_marginBottom="448dp"
        android:text="Welcome to My App!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

UI:



Lab 2: INTRODUCTION TO ACTIVITY AND LAYOUTS IN ANDROID

Q1: Create an app that illustrates that the activity lifecycle method being triggered by various action. Understand when onCreate(),onStart(),onResume event occur

MainActivity.java:

```
package com.example.demo;

import android.os.Bundle;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    Toast.makeText(this, "onCreate() called", Toast.LENGTH_SHORT).show();
}

@Override
protected void onStart() {
    super.onStart();
    Toast.makeText(this, "onStart() called", Toast.LENGTH_SHORT).show();
}

@Override
protected void onResume() {
    super.onResume();
    Toast.makeText(this, "onResume() called", Toast.LENGTH_SHORT).show();
}
}

```

activity_main.xml:

```

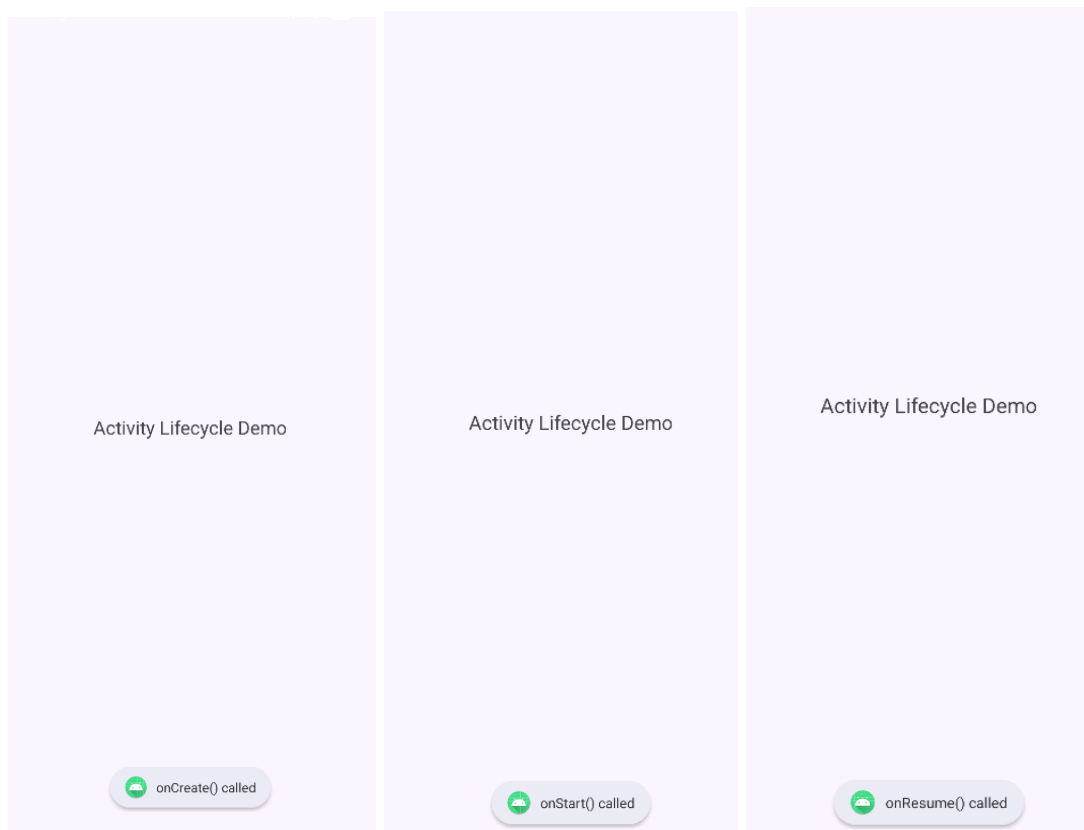
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity Lifecycle Demo"
        android:textSize="20sp"/>

</LinearLayout>

```

UI:



Q2: Create a Calculator app that does the function of multiplication, addition, division, subtraction but displays the result in the format of:-Num1 operator num2 = result. Back button on the next activity should get back to the calculator activity again.

MainActivity.java:

```
package com.example.demo;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

import com.example.demo.ResultActivity;

public class MainActivity extends AppCompatActivity {

    TextView display;
    String num1 = "", num2 = "", operator = "";
    boolean isSecondNumber = false;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        display = findViewById(R.id.display);
    }

    public void onDigitClick(View view) {
        Button btn = (Button) view;
        if (!isSecondNumber) {
            num1 += btn.getText().toString();
            display.setText(num1);
        } else {
            num2 += btn.getText().toString();
            display.setText(num2);
        }
    }

    public void onOperatorClick(View view) {
        Button btn = (Button) view;
        operator = btn.getText().toString();
        isSecondNumber = true;
    }
}
```



```

public void onClearClick(View view) {
    num1 = num2 = operator = "";
    isSecondNumber = false;
    display.setText("0");
}

public void onEqualClick(View view) {
    if (num1.isEmpty() || num2.isEmpty()) return;

    Intent intent = new Intent(this, ResultActivity.class);
    intent.putExtra("num1", num1);
    intent.putExtra("num2", num2);
    intent.putExtra("operator", operator);
    startActivity(intent);
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="12dp">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Calculator"
        android:textAlignment="center"
        android:textSize="30sp"/>

    <TextView
        android:id="@+id/display"
        android:layout_width="match_parent"
        android:layout_height="50dp"
        android:background="#DDDDDD"
        android:gravity="end|center_vertical"
        android:padding="10dp"
        android:textSize="28sp"

```

```
android:text="0"/>
```

```
<GridLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:columnCount="4"  
    android:rowCount="5"  
    android:layout_marginTop="10dp">
```

```
<Button android:text="7" android:onClick="onDigitClick"/>  
<Button android:text="8" android:onClick="onDigitClick"/>  
<Button android:text="9" android:onClick="onDigitClick"/>  
<Button android:text="÷" android:onClick="onOperatorClick"/>
```

```
<Button android:text="4" android:onClick="onDigitClick"/>  
<Button android:text="5" android:onClick="onDigitClick"/>  
<Button android:text="6" android:onClick="onDigitClick"/>  
<Button android:text="x" android:onClick="onOperatorClick"/>
```

```
<Button android:text="1" android:onClick="onDigitClick"/>  
<Button android:text="2" android:onClick="onDigitClick"/>  
<Button android:text="3" android:onClick="onDigitClick"/>  
<Button android:text="-" android:onClick="onOperatorClick"/>
```

```
<Button android:text="0" android:onClick="onDigitClick"/>  
<Button android:text="C" android:onClick="onClearClick"/>  
<Button android:text="=" android:onClick="onEqualClick"/>  
<Button android:text="+" android:onClick="onOperatorClick"/>
```

```
</GridLayout>
```

```
</LinearLayout>
```

ResultActivity.java:

```
package com.example.demo;
```

```

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class ResultActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_result);

        TextView resultText = findViewById(R.id.resultText);

        double n1 = Double.parseDouble(getIntent().getStringExtra("num1"));
        double n2 = Double.parseDouble(getIntent().getStringExtra("num2"));
        String op = getIntent().getStringExtra("operator");

        double result = 0;
        switch (op) {
            case "+": result = n1 + n2; break;
            case "-": result = n1 - n2; break;
            case "×": result = n1 * n2; break;
            case "÷": result = n2 != 0 ? n1 / n2 : 0; break;
        }

        resultText.setText(n1 + " " + op + " " + n2 + " = " + result);
    }
}

```

activity_result.xml:

```

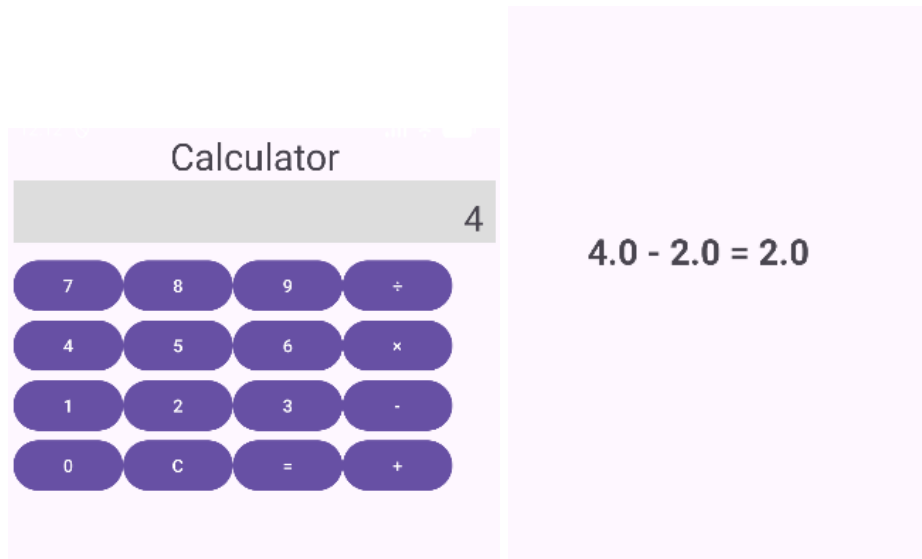
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/resultText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="26sp"
        android:textStyle="bold"/>

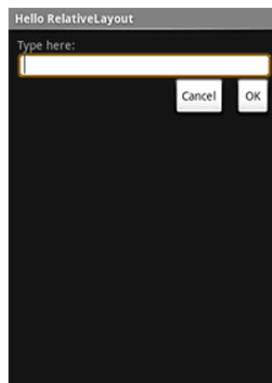
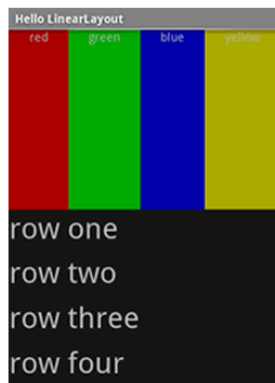
```

</LinearLayout>

UI:



Q3: Create the following given scenario using linear and relative layout concept.



Linear Layout:

MainActivity.java:

```
package com.example.demo;
```

```
import android.os.Bundle;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Hello Linear Layout"
        android:textSize="20sp"
        android:background="#8C000000"
        android:textColor="@color/white" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="281dp"
        android:orientation="horizontal">

        <TextView
            android:layout_width="0dp"
            android:layout_height="match_parent"
            android:layout_weight="1"
            android:background="#FF0000"
            android:textAlignment="center"
            android:text="red"
            android:textColor="#FFFFFF" />

        <TextView
            android:layout_width="0dp"
            android:layout_height="match_parent"
            android:layout_weight="1"
            android:background="#00FF00"
            android:textAlignment="center"

```

```
    android:text="green"  
    android:textColor="#FFFFFF" />
```

```
<TextView  
    android:layout_width="0dp"  
    android:layout_height="match_parent"  
    android:layout_weight="1"  
    android:background="#0000FF"  
    android:textAlignment="center"  
    android:text="blue"  
    android:textColor="#FFFFFF" />
```

```
<TextView  
    android:layout_width="0dp"  
    android:layout_height="match_parent"  
    android:layout_weight="1"  
    android:background="#FFFF00"  
    android:textAlignment="center"  
    android:text="yellow"  
    android:textColor="#FFFFFF" />
```

```
</LinearLayout>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:background="#222222"  
    android:orientation="vertical"  
    android:padding="16dp">
```

```
<TextView  
    android:layout_width="match_parent"  
    android:layout_height="0dp"  
    android:layout_weight="1"  
    android:text="row one"  
    android:textSize="22sp"  
    android:textColor="#FFFFFF"/>
```

```
<TextView  
    android:layout_width="match_parent"  
    android:layout_height="0dp"  
    android:layout_weight="1"  
    android:text="row two"  
    android:textSize="22sp"
```

```
android:textColor="#FFFFFF"/>
```

```
<TextView  
    android:layout_width="match_parent"  
    android:layout_height="0dp"  
    android:layout_weight="1"  
    android:text="row three"  
    android:textSize="22sp"  
    android:textColor="#FFFFFF"/>
```

```
<TextView  
    android:layout_width="match_parent"  
    android:layout_height="0dp"  
    android:layout_weight="1"  
    android:text="row four"  
    android:textSize="22sp"  
    android:textColor="#FFFFFF"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

UI :



Relative Layout:

MainActivity.java:

```
package com.example.demo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#111111"
    >

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Hello Relative Layout"
        android:background="#69767474"
        android:textColor="@color/white"
        />

    <TextView
        android:id="@+id/label"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Type here:"
```



```
android:textColor="#FFFFFF"
android:paddingLeft="16dp"
android:layout_marginTop="25dp"/>
```

```
<EditText
    android:id="@+id/inputBox"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/label"
    android:layout_marginTop="8dp"

    android:background="@drawable/input_border"/>
```

```
<Button
    android:id="@+id/cancelBtn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/inputBox"
    android:layout_alignParentEnd="true"
    android:layout_marginTop="14dp"
    android:layout_marginEnd="121dp"
    android:background="@drawable/flat_button"
    android:text="Cancel"
    android:textColor="@color/black" />
```

```
<Button
    android:id="@+id/okBtn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="OK"
    android:background="@drawable/flat_button"
    android:textColor="@color/black"
    android:layout_below="@id/inputBox"
    android:layout_alignParentEnd="true"
    android:layout_marginTop="12dp"
/>
```

```
</RelativeLayout>
```

input_border.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#FFFFFF"/>
```

```

    <corners android:radius="8dp"/>
    <stroke android:width="2dp" android:color="#FFD700"/>
    <padding android:left="8dp" android:right="8dp" android:top="8dp" android:bottom="8dp"/>
</shape>

```

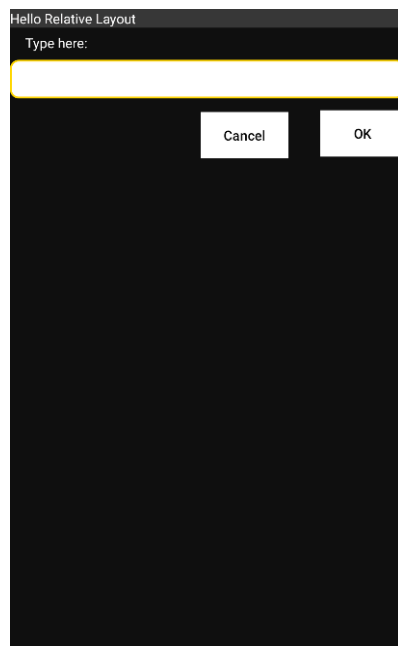
flat_button.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#E0E0E0"/>
    <stroke android:width="1dp" android:color="#888888"/>
    <corners android:radius="0dp"/>
</shape>

```

UI :



Q4: Create an app such that when the user click on the given URL typed by the user, it visits the corresponding page.

MainActivity.java:

```

package com.example.demo;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;

```

```

import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText urlInput;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        urlInput = findViewById(R.id.urlInput);
    }

    public void openWebsite(View view) {
        String url = urlInput.getText().toString().trim();

        if (url.isEmpty()) {
            Toast.makeText(this, "Please enter a URL", Toast.LENGTH_SHORT).show();
            return;
        }

        if (!url.startsWith("http://") && !url.startsWith("https://")) {
            url = "https://" + url;
        }

        Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse(url));
        startActivity(intent);
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

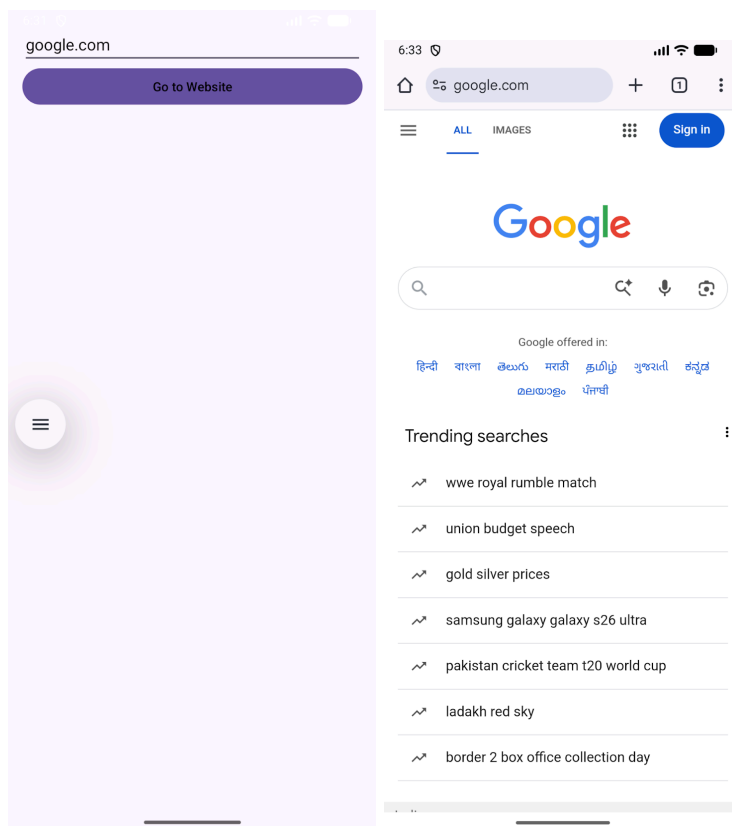
```

```
<EditText
    android:id="@+id/urlInput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter URL (e.g. google.com)"
    android:inputType="textUri"/>
```

```
<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Go to Website"
    android:textColor="@color/black"
    android:onClick="openWebsite"/>
```

```
</LinearLayout>
```

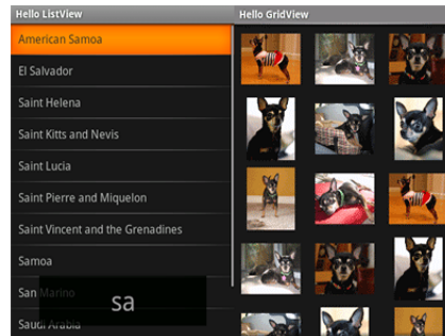
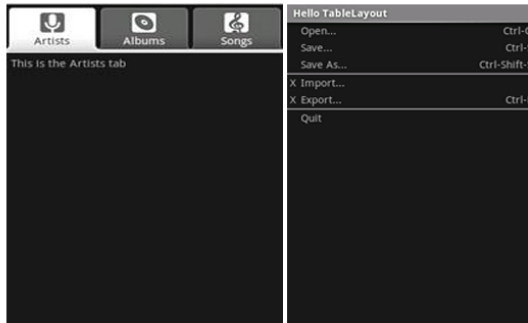
UI:



Lab 3: ACTIVITY AND LAYOUT CONTINUED

Q1: Using given scenario develop an application to perform following layout operations:

- List view ▪ Grid view ▪ Tab layout ▪ Table layout



MainActivity.java:

```
package com.example.demo;
```

```
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.viewpager2.widget.ViewPager2;
import com.google.android.material.tabs.TabLayout;
import com.google.android.material.tabs.TabLayoutMediator;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
        TabLayout tabLayout = findViewById(R.id.tabLayout);
        ViewPager2 viewPager = findViewById(R.id.viewPager);
```

```
        viewPager.setAdapter(new ViewPagerAdapter(this));
```

```
        new TabLayoutMediator(tabLayout, viewPager, (tab, position) -> {
            if (position == 0) tab.setText("Artists");
            else if (position == 1) tab.setText("Albums");
            else tab.setText("Songs");
        }).attach();
```

```
    }
}
```

AlbumsFragment.java:

```
package com.example.demo;

import android.os.Bundle;
import android.view.*;
import android.widget.GridView;
import androidx.fragment.app.Fragment;

public class AlbumsFragment extends Fragment {

    int[] images = {
        R.drawable.dog1, R.drawable.dog2, R.drawable.dog3,
        R.drawable.dog4, R.drawable.dog5, R.drawable.dog6
    };

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
        View v = inflater.inflate(R.layout.fragment_albums, container, false);
        GridView gridView = v.findViewById(R.id.gridView);
        gridView.setAdapter(new ImageAdapter(getContext(), images));
        return v;
    }
}
```

ArtistsFragment.java:

```
package com.example.demo;

import android.os.Bundle;
import android.view.*;
import android.widget.*;
import androidx.fragment.app.Fragment;

public class ArtistsFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
        View v = inflater.inflate(R.layout.fragment_artists, container, false);

        String[] countries = {
            "American Samoa", "El Salvador", "Saint Helena",
            "Saint Kitts and Nevis", "Saint Lucia", "Saint Pierre and Miquelon",
            "Saint Vincent and the Grenadines", "Samoa", "San Marino", "Saudi Arabia"
        };
    }
}
```

```

};

ListView listView = v.findViewById(R.id.listView);
ArrayAdapter<String> adapter = new ArrayAdapter<>(getContext(),
    R.layout.list_item_white,
    R.id.itemText,
    countries);

listView.setAdapter(adapter);

return v;
}
}

```

ImageAdapter.java:

```

package com.example.demo;

import android.content.Context;
import android.view.*;
import android.widget.*;

public class ImageAdapter extends BaseAdapter {

    Context context;
    int[] images;

    public ImageAdapter(Context c, int[] imgs) {
        context = c;
        images = imgs;
    }

    public int getCount() { return images.length; }
    public Object getItem(int pos) { return images[pos]; }
    public long getItemId(int pos) { return pos; }

    public View getView(int pos, View convertView, ViewGroup parent) {
        ImageView iv = new ImageView(context);
        iv.setImageResource(images[pos]);
        iv.setLayoutParams(new GridView.LayoutParams(300, 300));
        iv.setScaleType(ImageView.ScaleType.CENTER_CROP);
        return iv;
    }
}

```

SongsFragment.java:

```
package com.example.demo;

import android.os.Bundle;
import android.view.*;
import androidx.fragment.app.Fragment;

public class SongsFragment extends Fragment {
    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
        return inflater.inflate(R.layout.fragment_songs, container, false);
    }
}
```

ViewPagerAdapter.java:

```
package com.example.demo;

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentActivity;
import androidx.viewpager2.adapter.FragmentStateAdapter;

public class ViewPagerAdapter extends FragmentStateAdapter {

    public ViewPagerAdapter(@NonNull FragmentActivity fa) {
        super(fa);
    }

    @NonNull
    @Override
    public Fragment createFragment(int position) {
        if (position == 0) return new ArtistsFragment();
        else if (position == 1) return new AlbumsFragment();
        else return new SongsFragment();
    }

    @Override
    public int getItemCount() {
        return 3;
    }
}
```


activity_main.xml:

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <com.google.android.material.tabs.TabLayout
        android:id="@+id/tabLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="#333333"
        app:tabTextColor="@color/white"
        app:tabSelectedTextColor="#FFFFFF"
        app:tabIndicatorColor="#FFFFFF"/>

    <androidx.viewpager2.widget.ViewPager2
        android:id="@+id/viewPager"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
    />
</LinearLayout>
```

fragment_albums.xml:

```
<GridView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/gridView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:numColumns="3"
    android:verticalSpacing="8dp"
    android:horizontalSpacing="8dp"
    android:background="#111111"
    android:padding="8dp"/>
```

fragment_artists.xml:

```
<ListView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/listView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#111111"
    android:divider="#444444"
    android:dividerHeight="1dp"/>
```

fragment_songs.xml:

```
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#111111"
    android:stretchColumns="*"
    android:padding="10dp">

    <TableRow>
        <TextView android:text="Open..." android:textColor="#FFFFFF"/>
        <TextView android:text="Ctrl+O" android:textColor="#AAAAAA"/>
    </TableRow>

    <TableRow>
        <TextView android:text="Save..." android:textColor="#FFFFFF"/>
        <TextView android:text="Ctrl+S" android:textColor="#AAAAAA"/>
    </TableRow>

    <TableRow>
        <TextView android:text="Import..." android:textColor="#FFFFFF"/>
        <TextView android:text="Ctrl+I" android:textColor="#AAAAAA"/>
    </TableRow>

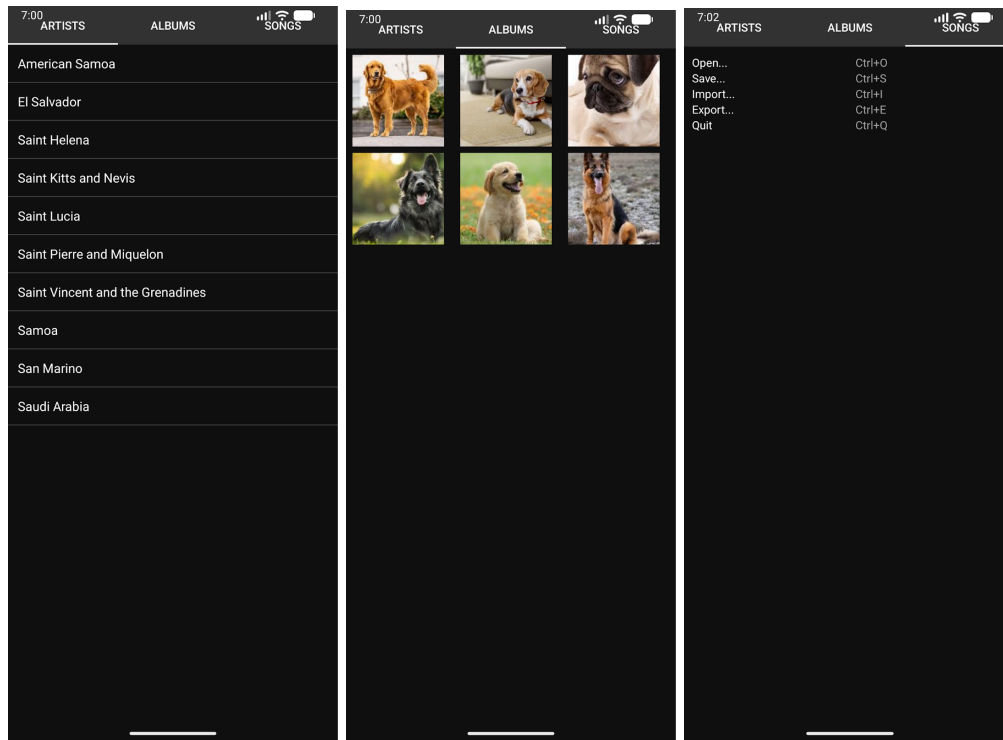
    <TableRow>
        <TextView android:text="Export..." android:textColor="#FFFFFF"/>
        <TextView android:text="Ctrl+E" android:textColor="#AAAAAA"/>
    </TableRow>

    <TableRow>
        <TextView android:text="Quit" android:textColor="#FFFFFF"/>
        <TextView android:text="Ctrl+Q" android:textColor="#AAAAAA"/>
    </TableRow>
</TableLayout>
```

list_item_white.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/itemText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textColor="#FFFFFF"
    android:textSize="16sp"
    android:padding="12dp"
    android:background="#111111"/>
```

UI:



Q2: Write a program to list different sports, and when a sport is selected, display a message showing the selected sport.

MainActivity.java:

```
package com.example.demo;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ListView;
import android.widget.Toast;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    ListView sportsListView;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

sportsListView = findViewById(R.id.sportsListView);
```

```
String[] sports = {
    "Cricket",
    "Football",
    "Tennis",
    "Basketball",
    "Hockey",
    "Badminton",
    "Volleyball"
};
```

```
ArrayAdapter<String> adapter = new ArrayAdapter<>(
    this,
    android.R.layout.simple_list_item_1,
    sports
);
```

```
sportsListView.setAdapter(adapter);
```

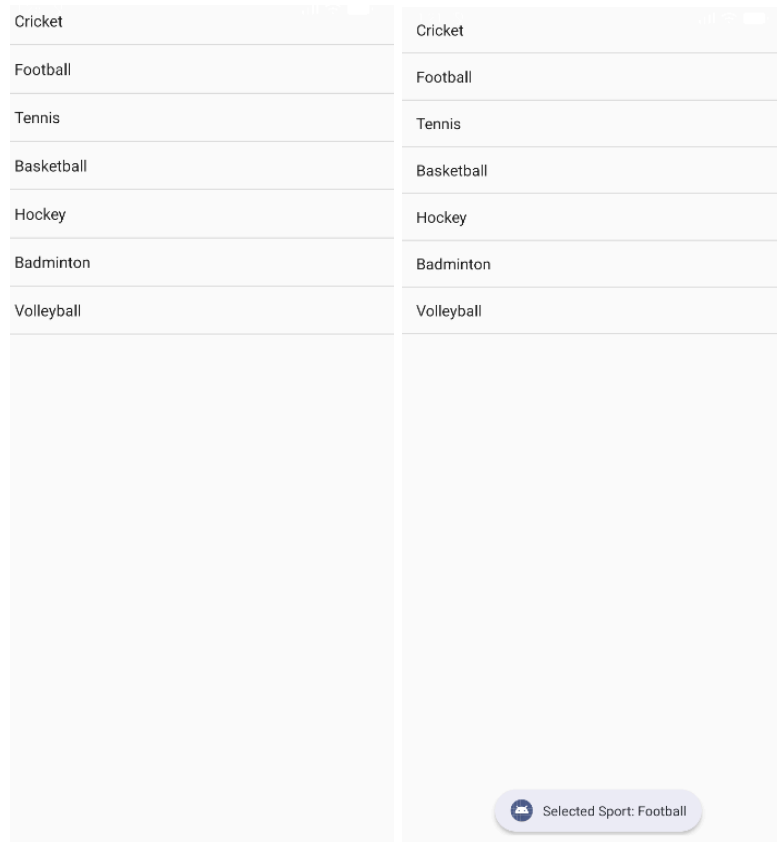
```
sportsListView.setOnItemClickListener((parent, view, position, id) -> {
    String selectedSport = sports[position];
    Toast.makeText(MainActivity.this,
        "Selected Sport: " + selectedSport,
        Toast.LENGTH_SHORT).show();
    });
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<ListView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/sportsListView"
```

```
android:layout_width="match_parent"
android:layout_height="match_parent" />
```

UI:



Q3: Design an news application and Implement the navigation between sections like Top Stories, Sports, and Entertainment using tab layout.

MainActivity.java:

```
package com.example.demo;
```

```
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.viewpager.widget.ViewPager;
import com.google.android.material.tabs.TabLayout;
```

```
public class MainActivity extends AppCompatActivity {
```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    TabLayout tabLayout = findViewById(R.id.tabLayout);
    ViewPager viewPager = findViewById(R.id.viewPager);

    ViewPagerAdapter adapter = new ViewPagerAdapter(getSupportFragmentManager());
    adapter.addFragment(new TopStoriesFragment(), "Top Stories");
    adapter.addFragment(new SportsFragment(), "Sports");
    adapter.addFragment(new EntertainmentFragment(), "Entertainment");

    viewPager.setAdapter(adapter);
    tabLayout.setupWithViewPager(viewPager);
}
}

```

EntertainmentFragment.java:

```

package com.example.demo;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ArrayAdapter;
import android.widget.ListView;

import androidx.fragment.app.Fragment;

public class EntertainmentFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
        View v = inflater.inflate(R.layout.fragment_list, container, false);
        ListView listView = v.findViewById(R.id.listView);

        String[] entertainmentNews = {
            "New Movie Releases",
            "Celebrity Interviews",

```

```

        "Music Festival Updates",
        "Award Show Highlights"
    };

    ArrayAdapter<String> adapter = new ArrayAdapter<>(getContext(),
        android.R.layout.simple_list_item_1, entertainmentNews);

    listView.setAdapter(adapter);
    return v;
}
}

```

SportsFragment.java:

```

package com.example.demo;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ArrayAdapter;
import android.widget.ListView;

import androidx.fragment.app.Fragment;

public class SportsFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
        View v = inflater.inflate(R.layout.fragment_list, container, false);
        ListView listView = v.findViewById(R.id.listView);

        String[] sportsNews = {
            "Football World Cup Updates",
            "Cricket Series Highlights",
            "Olympics Preparation",
            "Tennis Grand Slam News"
        };

        ArrayAdapter<String> adapter = new ArrayAdapter<>(getContext(),
            android.R.layout.simple_list_item_1, sportsNews);

        listView.setAdapter(adapter);
        return v;
    }
}

```

```
}  
}
```

TopStoriesFragment.java:

```
package com.example.demo;
```

```
import android.os.Bundle;  
import android.view.LayoutInflater;  
import android.view.View;  
import android.view.ViewGroup;  
import android.widget.AdapterView;  
import android.widget.AdapterView.OnItemClickListener;  
import android.widget.ListView;
```

```
import androidx.fragment.app.Fragment;
```

```
public class TopStoriesFragment extends Fragment {
```

```
    @Override
```

```
    public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle  
savedInstanceState) {
```

```
        View v = inflater.inflate(R.layout.fragment_list, container, false);  
        ListView listView = v.findViewById(R.id.listView);
```

```
        String[] news = {  
            "Global Market Updates",  
            "New Technology Trends",  
            "Political Developments",  
            "Climate Change News"  
        };
```

```
        ArrayAdapter<String> adapter = new ArrayAdapter<>(getContext(),  
            android.R.layout.simple_list_item_1, news);
```

```
        listView.setAdapter(adapter);  
        return v;
```

```
    }  
}
```


ViewPagerAdapter.java:

```
package com.example.demo;

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentPagerAdapter;
import java.util.ArrayList;
import java.util.List;

public class ViewPagerAdapter extends FragmentPagerAdapter {

    private final List<Fragment> fragmentList = new ArrayList<>();
    private final List<String> fragmentTitleList = new ArrayList<>();

    public ViewPagerAdapter(FragmentManager manager) {
        super(manager);
    }

    @NonNull
    @Override
    public Fragment getItem(int position) {
        return fragmentList.get(position);
    }

    @Override
    public int getCount() {
        return fragmentList.size();
    }

    public void addFragment(Fragment fragment, String title) {
        fragmentList.add(fragment);
        fragmentTitleList.add(title);
    }

    @Override
    public CharSequence getPageTitle(int position) {
        return fragmentTitleList.get(position);
    }
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <com.google.android.material.tabs.TabLayout
        android:id="@+id/tabLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <androidx.viewpager.widget.ViewPager
        android:id="@+id/viewPager"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

fragment_list.xml:

```
<ListView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/listView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

UI:

TOP STORIES	SPORTS	ENTERTAINMENT
Global Market Updates	Football World Cup Updates	
New Technology Trends	Cricket Series Highlights	
Political Developments	Olympics Preparation	
Climate Change News	Tennis Grand Slam News	

TOP STORIES	SPORTS	ENTERTAINMENT
		New Movie Releases
		Celebrity Interviews
		Music Festival Updates
		Award Show Highlights

Lab 4: INPUT CONTROLS IN ANDROID

Q1: Develop a "Test App" that includes a layout with a Button and a ToggleButton. When each button is clicked, a custom Toast message should be displayed with different images as their content.

MainActivity.java:

```
package com.example.demo;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import android.widget.ToggleButton;

import androidx.appcompat.app.AppCompatActivity;
```

```

public class MainActivity extends AppCompatActivity {

    Button btnShow;
    ToggleButton toggleShow;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnShow = findViewById(R.id.btnShow);
        toggleShow = findViewById(R.id.toggleShow);

        btnShow.setOnClickListener(v -> showCustomToast("Button Clicked!",
R.drawable.button_image));

        toggleShow.setOnCheckedChangeListener((buttonView, isChecked) -> {
            if (isChecked) {
                showCustomToast("Toggle ON", R.drawable.toggle_on);
            } else {
                showCustomToast("Toggle OFF", R.drawable.toggle_off);
            }
        });
    }

    private void showCustomToast(String message, int imageRes) {
        LayoutInflater inflater = getLayoutInflater();
        View layout = inflater.inflate(R.layout.custom_toast, null);

        ImageView image = layout.findViewById(R.id.toastImage);
        TextView text = layout.findViewById(R.id.toastText);

        image.setImageResource(imageRes);
        text.setText(message);

        Toast toast = new Toast(getApplicationContext());
        toast.setDuration(Toast.LENGTH_SHORT);
        toast.setView(layout);
        toast.show();
    }
}

```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <Button
        android:id="@+id/btnShow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Toast" />

    <ToggleButton
        android:id="@+id/toggleShow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="Toast ON"
        android:textOff="Toast OFF"
        android:layout_marginTop="20dp" />

</LinearLayout>
```

custom_toast.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="10dp"
    android:background="#DD000000"
    android:gravity="center">

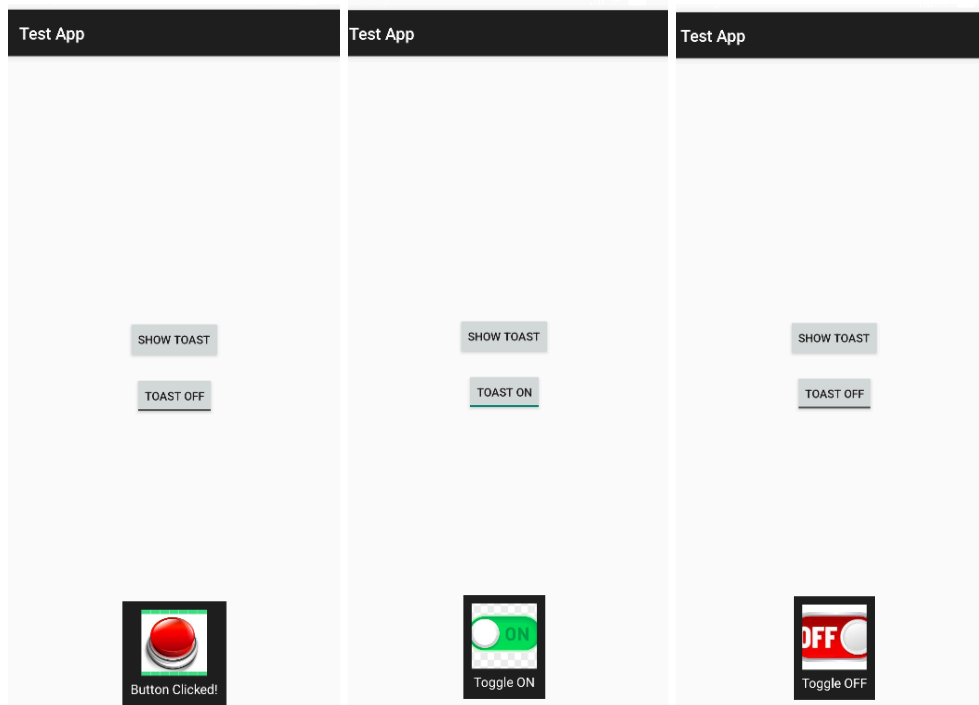
    <ImageView
        android:id="@+id/toastImage"
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:src="@drawable/ic_launcher_foreground" />
```

```

<TextView
    android:id="@+id/toastText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textColor="#FFFFFF"
    android:textSize="16sp"
    android:paddingTop="5dp"/>
</LinearLayout>

```

UI:



Q2: Create an app that contains a view with multiple buttons, each labeled with different Android versions. When a button is clicked, a Toast message should appear, displaying the corresponding Android version's name along with its associated icon.

MainActivity.java:

```

package com.example.demo;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;

```

```

import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        findViewById(R.id.btnCupcake).setOnClickListener(v ->
            showToast("Android Cupcake", R.drawable.cupcake));

        findViewById(R.id.btnKitKat).setOnClickListener(v ->
            showToast("Android KitKat", R.drawable.kitkat));

        findViewById(R.id.btnLollipop).setOnClickListener(v ->
            showToast("Android Lollipop", R.drawable.lollipop));

        findViewById(R.id.btnPie).setOnClickListener(v ->
            showToast("Android Pie", R.drawable.pie));
    }

    private void showToast(String message, int imageRes) {
        LayoutInflater inflater = getLayoutInflater();
        View layout = inflater.inflate(R.layout.custom_toast, null);

        ImageView image = layout.findViewById(R.id.toastImage);
        TextView text = layout.findViewById(R.id.toastText);

        image.setImageResource(imageRes);
        text.setText(message);

        Toast toast = new Toast(getApplicationContext());
        toast.setDuration(Toast.LENGTH_SHORT);
        toast.setView(layout);
        toast.show();
    }
}

```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="20dp">

        <Button
            android:id="@+id/btnCupcake"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Android 1.5 - Cupcake"/>

        <Button
            android:id="@+id/btnKitKat"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Android 4.4 - KitKat"
            android:layout_marginTop="10dp"/>

        <Button
            android:id="@+id/btnLollipop"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Android 5.0 - Lollipop"
            android:layout_marginTop="10dp"/>

        <Button
            android:id="@+id/btnPie"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Android 9 - Pie"
            android:layout_marginTop="10dp"/>

    </LinearLayout>

</ScrollView>
```



```
</LinearLayout>
</ScrollView>
```

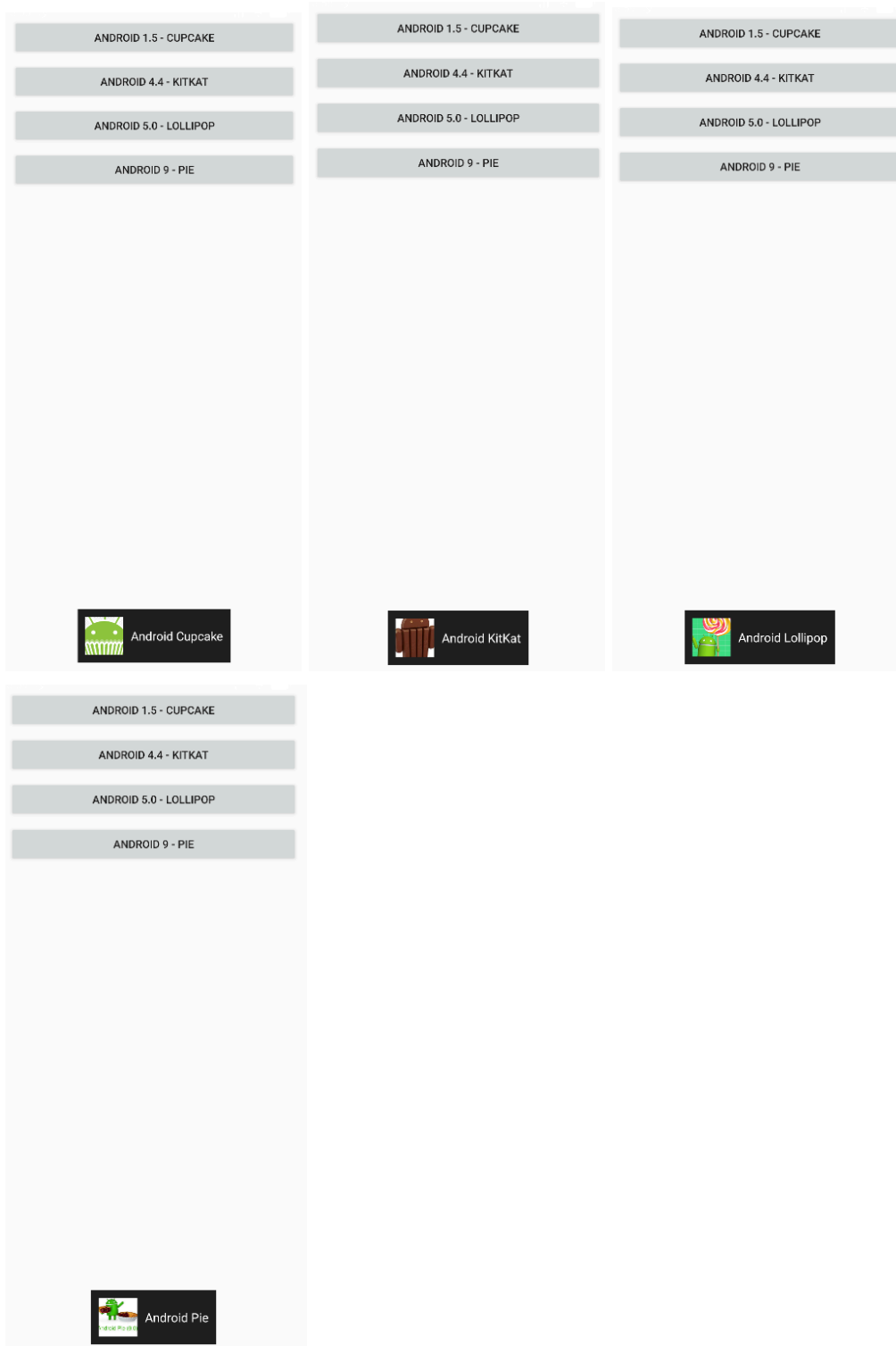
custom_toast.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:padding="10dp"
    android:background="#DD000000"
    android:gravity="center_vertical">

    <ImageView
        android:id="@+id/toastImage"
        android:layout_width="50dp"
        android:layout_height="50dp"
        android:layout_marginEnd="10dp"/>

    <TextView
        android:id="@+id/toastText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textColor="#FFFFFF"
        android:textSize="16sp"/>
</LinearLayout>
```

UI:



Q3: Develop a view with a ToggleButton labeled "Current Mode" that has two states: "Wi-Fi" and "Mobile Data." Based on the state of the toggle button, an image corresponding to the selected mode should appear, and a Toast message should

display the current mode. Additionally, when the user clicks the "Change Mode" button, the app should switch to the corresponding mode and update the image accordingly.

MainActivity.java:

```
package com.example.demo;

import android.os.Bundle;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.Toast;
import android.widget.ToggleButton;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ToggleButton toggleMode;
    ImageView modelImage;
    Button btnChange;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        toggleMode = findViewById(R.id.toggleMode);
        modelImage = findViewById(R.id.modelImage);
        btnChange = findViewById(R.id.btnChange);

        toggleMode.setOnCheckedChangeListener((buttonView, isChecked) -> {
            if (isChecked) {
                modelImage.setImageResource(R.drawable.wifi);
                Toast.makeText(this, "Current Mode: Wi-Fi", Toast.LENGTH_SHORT).show();
            } else {
                modelImage.setImageResource(R.drawable.mobile_data);
                Toast.makeText(this, "Current Mode: Mobile Data", Toast.LENGTH_SHORT).show();
            }
        });

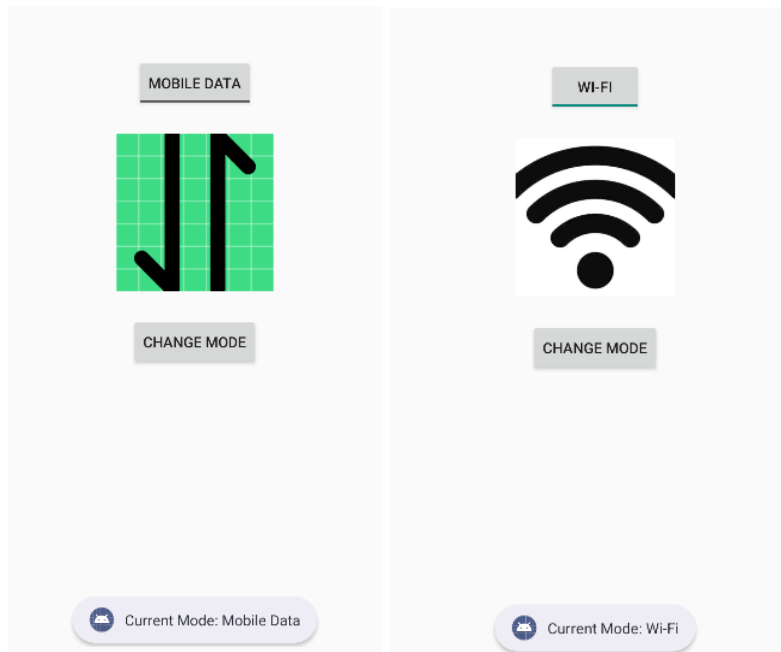
        btnChange.setOnClickListener(v -> {
            toggleMode.toggle();
        });
    }
}
```

```
}  
}
```

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>  
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="vertical"  
    android:gravity="center"  
    android:padding="24dp">  
  
    <ToggleButton  
        android:id="@+id/toggleMode"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:textOn="Wi-Fi"  
        android:textOff="Mobile Data"  
        android:text="Current Mode" />  
  
    <ImageView  
        android:id="@+id/modelImage"  
        android:layout_width="150dp"  
        android:layout_height="150dp"  
        android:layout_marginTop="24dp"  
        android:src="@drawable/mobile_data" />  
  
    <Button  
        android:id="@+id/btnChange"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:text="Change Mode"  
        android:layout_marginTop="24dp"/>  
  
</LinearLayout>
```

UI:



Q4: Create a “Food Ordering App” which lists food items with check boxes. Once the user checks /unchecks the item and click on the submit button display the items ordered along with cost of each item and total cost in a new activity. Once the user clicks on the submit button he/she should not be allowed to change the order i.e. he should not be allowed to change the state of the items checked.

MainActivity.java:

```
package com.example.demo;
```

```
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    CheckBox cbBurger, cbPizza, cbPasta, cbSandwich, cbCoffee;
    Button btnSubmit;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```

super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

cbBurger = findViewById(R.id.cbBurger);
cbPizza = findViewById(R.id.cbPizza);
cbPasta = findViewById(R.id.cbPasta);
cbSandwich = findViewById(R.id.cbSandwich);
cbCoffee = findViewById(R.id.cbCoffee);
btnSubmit = findViewById(R.id.btnSubmit);

btnSubmit.setOnClickListener(v -> {

    StringBuilder orderDetails = new StringBuilder();
    int total = 0;

    if (cbBurger.isChecked()) {
        orderDetails.append("Burger - ₹120\n");
        total += 120;
    }
    if (cbPizza.isChecked()) {
        orderDetails.append("Pizza - ₹250\n");
        total += 250;
    }
    if (cbPasta.isChecked()) {
        orderDetails.append("Pasta - ₹180\n");
        total += 180;
    }
    if (cbSandwich.isChecked()) {
        orderDetails.append("Sandwich - ₹100\n");
        total += 100;
    }
    if (cbCoffee.isChecked()) {
        orderDetails.append("Coffee - ₹80\n");
        total += 80;
    }

    Intent intent = new Intent(MainActivity.this, ResultActivity.class);
    intent.putExtra("order", orderDetails.toString());
    intent.putExtra("total", total);
    startActivity(intent);

    cbBurger.setEnabled(false);
    cbPizza.setEnabled(false);

```

```

        cbPasta.setEnabled(false);
        cbSandwich.setEnabled(false);
        cbCoffee.setEnabled(false);
        btnSubmit.setEnabled(false);
    });
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:text="Select Your Food"
            android:textSize="22sp"
            android:textStyle="bold"
            android:paddingBottom="16dp" android:layout_height="wrap_content"
            android:layout_width="wrap_content"/>

        <CheckBox
            android:id="@+id/cbBurger"
            android:text="Burger - ₹120" android:layout_height="wrap_content"
            android:layout_width="wrap_content"/>

        <CheckBox
            android:id="@+id/cbPizza"
            android:text="Pizza - ₹250" android:layout_height="wrap_content"
            android:layout_width="wrap_content"/>

        <CheckBox
            android:id="@+id/cbPasta"
            android:text="Pasta - ₹180" android:layout_height="wrap_content"
            android:layout_width="wrap_content"/>

        <CheckBox

```

```
        android:id="@+id/cbSandwich"
        android:text="Sandwich - ₹100" android:layout_height="wrap_content"
    android:layout_width="wrap_content"/>
```

```
<CheckBox
    android:id="@+id/cbCoffee"
    android:text="Coffee - ₹80" android:layout_height="wrap_content"
    android:layout_width="wrap_content"/>
```

```
<Button
    android:id="@+id/btnSubmit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Submit Order"
    android:layout_marginTop="20dp"/>
```

```
</LinearLayout>
</ScrollView>
```

ResultActivity.java:

```
package com.example.demo;
```

```
import android.os.Bundle;
import android.widget.TextView;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class ResultActivity extends AppCompatActivity {
```

```
    TextView tvOrder, tvTotal;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_result);
```

```
        tvOrder = findViewById(R.id.tvOrder);
        tvTotal = findViewById(R.id.tvTotal);
```

```
        String order = getIntent().getStringExtra("order");
        int total = getIntent().getIntExtra("total", 0);
```

```
        tvOrder.setText(order);
        tvTotal.setText("Total Cost: ₹" + total);
```



```
}  
}
```

activity_result.xml:

```
<?xml version="1.0" encoding="utf-8"?>  
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="vertical"  
    android:padding="20dp">  
  
    <TextView  
        android:text="Your Order"  
        android:textSize="22sp"  
        android:textStyle="bold"  
        android:paddingBottom="16dp" android:layout_height="wrap_content"  
    android:layout_width="wrap_content"/>  
  
    <TextView  
        android:id="@+id/tvOrder"  
        android:textSize="18sp" android:layout_height="wrap_content"  
    android:layout_width="wrap_content"/>  
  
    <TextView  
        android:id="@+id/tvTotal"  
        android:textSize="20sp"  
        android:textStyle="bold"  
        android:layout_marginTop="20dp" android:layout_height="wrap_content"  
    android:layout_width="wrap_content"/>  
  
</LinearLayout>
```

UI:

Select Your Food ☐ Burger - ₹120 ☐ Pizza - ₹250 ☐ Pasta - ₹180 ☒ Sandwich - ₹100 ☒ Coffee - ₹80 SUBMIT ORDER	Your Order Sandwich - ₹100 Coffee - ₹80 Total Cost: ₹180
--	---

